Effect of ZrO_2 on copolymer based gel polymer electrolyte in Li-Secondary Batteries

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***Abstract*— Polymer electrolyte membranes that consists fixed ratio of PVdC- co-AN, $LiClO_4$, PC with different concentration of ZrO_2 have been prepared by solution casting technique. The structural studies are carried out by X-ray diffraction analysis. The formation of polymer-salt complexes has been confirmed by and FT-IR spectral studies. The as prepared electrolytes i.e., 3(S1), 6(S2) and 9 (S3) wt% of ZrO_2 based PVdC-co-AN(0.7)/$LiClO_4$(0.2)/PC(0.1) system has better complexation which was elucidated using XRD and FT-IR analyses. The conductivity increases with the increase of ZrO_2 content and reaches the maximum of 4.87×10^{-3} S/cm. This is higher than the bare one which was increased by two order when adding 9 wt% ZrO_2.**

Keywords—polymer complexes, conductivity, ZrO_2

I. INTRODUCTION

In the recent years R&D works based on the development of new salts for lithium ion batteries are of popular interest. Electrolyte is a key factor which limits the performance of the batteries. The developments of polymer electrolytes have become more commonly used in solid state generators with high energy density. Most of the procedures for preparing gel polymer electrolytes need a moisture-free atmosphere because of the higher water sensitivity of the lithium salt. Furthermore, their mechanical properties are often very poor [1]. Mucha and Kryszewski [2] described that polymer blends of PVdC and PAN begin to undergo destruction at lower temperatures than homopolymers PVdC and PAN. Among the polymer hosts studied, so far, the PAN based electrolytes offer a homogenous, hybrid electrolyte films in which the salt and the plasticizer were molecularly dispersed. Watanabe et al. [3,4] used a combination of plasticizers EC and PC to plasticize PAN and complexed with LiClO4 and they correlated with the mole ratio [plasticizer]/ [$LiClO_4$] of the gel electrolytes. The authors concluded that the PAN host is inactive in the ionic transport mechanism but acts as a matrix for structural stability. Over all, poly (vinylidene chloride-co-acrylonitrile) has drawn much attention of researchers, because it is a copolymer having reasonable thermal and mechanical properties.

Despite the numerous research works are going on at present, practical lithium batteries based on SPE have remained impalpable. Some of the problems that hindered the development of SPE include low conductivity at ambient temperature and their reactivity with the lithium metal electrode insolvent plasticized polymer systems [5,6]. The main difficulty to solve these problems is due to the ionic conductivity and mechanical strength of a polymer electrolyte being disparate to each other, i.e. mechanical strength of the electrolyte decreases as conductivity increases. In order to overcome thisobstacle, high surface area inorganic fillers such as Al_2O_3, ZrO_2, and TiO_2 are added into the polymers matrices and are called composite polymer electrolyte (CPE). The primary reasons to incorporate these fillers are twofold: (i) enhancement in ionic conductivity at low temperature and (ii) improvement of the stability at the interface with the electrode. Incorporation of the fillers will help to inhibit the recrystallization of the polymer and decrease the glass transition temperature of the CPE and hence enhance the conductivity [6].In the present work, a fixed ratio of of PVdC- co-AN, $LiClO_4$, PC with different concentration of ZrO_2 prepared by using solution casting technique.

II. EXPERIMENTAL

The specific amount of PVdC-co-AN with salt ($LiClO_4$) and propylene corbanate (PC) are weight using the electrolyte balance. The polymer and salt preheated at 100°C in a vacuum controlled oven to remove moisture present in the system. The solvent (THF) is used without any further purification. It was then allowed to cool inside the desicators after treating with the solvent THF. The mixture was stirred for about 24 hours at room temperature using .Then it was heated 60°C and stirrer one hour at the same tempera-

ture. The calculated amount of filler (ZrO_2) are dispersed with the homogeneous solution and stirred. The excess amount of solvent are evaporated and the solution in a flat bottom Petridis.

III. Results and Discussion

The ExPert Pro PAN alytical X-ray diffractometer was used to do the structural analysis of the films at room temperature. XRD shows (Fig.1) the diffraction patterns for PVdC-co-AN, $LiClO_4$ and all the prepared compositions of PVdC-co-AN/$LiClO_4$ polymer electrolytes. The characteristic peaks related to the crystalline phase of PVdC-co-AN diminish with the addition of $LiClO_4$ which indicates that the lithium salt is not crystallized in the film, but rather molecularly dispersed in PVdC-co-AN. Moreover, the crystalline peaks pertaining to $LiClO_4$ also disappear in the complexes which indicate the complete dissociation of the salt. This, in turn, reveals the better miscibility of the polymer and salt. The miscibility is nothing but the ability to be mixed at molecular level so that there is a good interaction between functional groups of polymer and salt that results new systems with improved properties. The diffraction peaks at $\Theta = 28.15°$ and $31.44°$ indicate the presence of ZrO_2. The intensity of the filler decreases with respect to the concentration of filler increases.

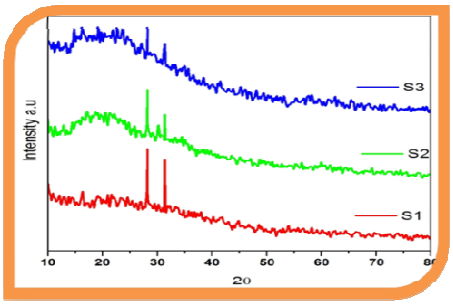

Fig.1.XRD patterns for (S1)PVdC-co-AN(0.7)/$LiClO_4$(0.2)/PC(0.1)/3wt% of ZrO_2(S2) PVdC-co-AN(0.7)/$LiClO_4$(0.2)/PC(0.1)/6wt% of ZrO_2(S3) PVdC-co-AN(0.7)/$LiClO_4$(0.2)/PC(0.1)/9wt% of ZrO_2 polymer complexes

Figure.2 illustrates the FTIR spectra of the prepared samples S1, S2 and S3 with different ratios of filler, and Table.1 shows the various band assignments corresponding to the wave numbers. The broad vibrational peak around 3,600 cm^{-1} the spectra pertaining to the –OH group [7] is attributed to the water or moisture absorption which might have happened during the sample loading for the spectral analysis. The well-built vibrations of C≡N and C=N bands in PVdC-co-AN are represented by the peaks around 2,245 cm^{-1} is well matched in all as prepared samples and 1,642 cm^{-1} is shifted to the wave numbers of 1,649 cm^{-1} and 1,660 cm^{-1} respectively in the S1, S2 and S3 gel polymer electrolytes. The disappearance of the few characteristic peaks of the plasticizers and the appearance of the new peaks and shift in the peaks confirm the complex formation between polymers, salt, and different ratios of fillers.

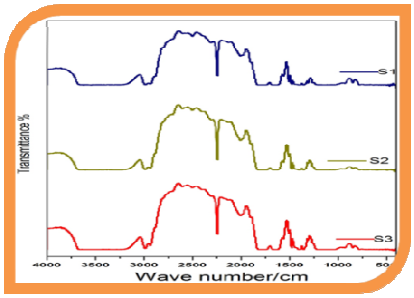

Fig.2.FTIR transmission spectra for S1,S2 and S3 polymer complexes

Band assignment	Wave number (cm^{-1})		
	S1	S2	S3
CH_2 symmetric stretching	2987	2986	2986
C=N stretching	1649	1660	1649
CH_2 scissoring	1483	1483	1482
CH_2 twisting	1357	1357	1357
CH_2 rocking	848	849	849

Table.1 Assignments of FTIR absorption bands for the prepared samples

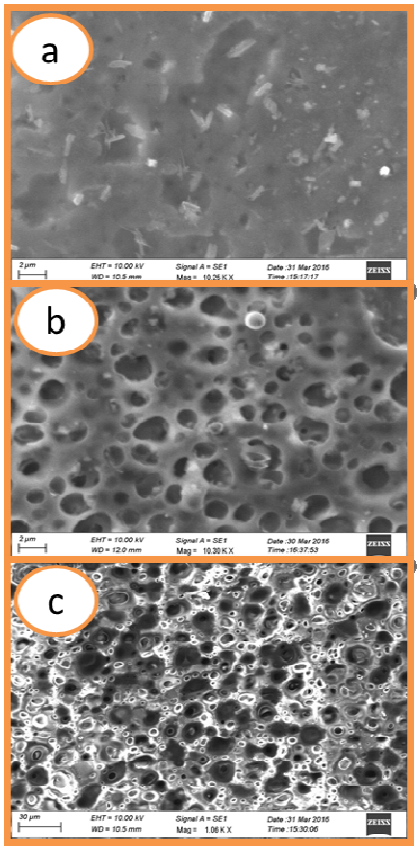

Fig.3.SEM images for the as prepaared polymer complexes

The scanning electron microscopic images of the prepared electrolytes were studied for its surface information with various magnification and shown in the Fig.3 (a-c). It is observed from the Fig.3 that the voids are created more and more upon adding ZrO_2 content which resembles the conductivity values significantly. The increase of pores and its sizes could be evidenced from the Fig.3

The ionic conductivity of polymer electrolytes was carried out with the help of stainless steel blocking electrodes using a computer controlled-Autolab Type-III Potentiostat/Galvanostat in the frequency range of 40 Hz to 300 kHz with the signal amplitude of 0.25 V. The measurements were taken between 303 and 353 K. The typical impedance plot for PVdC-co-AN gel polymer electrolytes at room temperature is presented in Fig.4(a-c). The highest ionic conductivity has been observed at 9 wt. % of filler in the overall range of temperatures studied. The ionic conductivities of PVdC-co-AN/$LiClO_4$ complexes with different ratios of ZrO_2 are shown in Figure 4(d). Arrhenius plots of 1000/T versus log conductivity of polymer complexes in the temperature range 303K – 353K From this, the activation energy values has been tabulated in the Table.2. Ionic conductivity as a function of plasticizer ratio can be generally explained by the decrease in glass transition temperature (Tg) competing with the increase in the number of charge carriers. As the filler ratio increases, the ionic conductivity also increases, which is due to the increment in the mobility of charge carriers in the polymer–salt complexes. The observed maximum ionic conductivity value shows a balance between number of charge carriers and ion mobility (8,9).The variations in ln with inverse of absolute temperature for all the prepared polymer electrolyte films are shown in Figure 3(d). From the figure, it is observed that the increase in temperature increases the conductivity of polymer electrolytes over the studied temperature range.

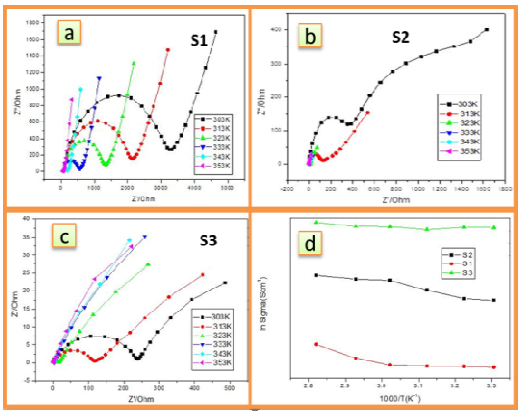

Fig.4(a-c) Impedance studies of polymer complexes(d) Plot of temperature versus log conductivity for S1,S2 and S3 samples

Sample code	RT	313K	323K	333K	343K	353K	Activation energy
S1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
S2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
S3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Table.2 ionic conductivity and activation energy of the polymer samples

IV. Conclusion

Polymer electrolyte membranes that consists fixed ratio of PVdC- co-AN, $LiClO_4$, PC with different concentration of ZrO_2 have been prepared by solution casting technique. The as prepared electrolytes i.e., 3(S1), 6(S2) and 9 (S3) wt% of ZrO_2 based PVdC-co-AN(0.7)/$LiClO_4$(0.2)/PC(0.1) system has better complexation which was elucidated using XRD and FT-IR analyses. The conductivity increases with the increase of ZrO_2 content and reaches the maximum of 4.87×10^{-3} S/cm. This is higher than the bare one which was increased by two order when adding 9 wt% ZrO_2. This was supported by the SEM analysis.

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